Cortex: Sleep-Inspired Memory Consolidation for Language Agents

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Abstract

Sleep-dependent memory research and complementary learning systems theory motivate a separation between rapid episodic capture and slower integration. Cortex uses this separation to preserve attributed observations while consolidating derived memories. A research prototype tests NREM-inspired source-grounded replay and REM-inspired recombination during classifier training. These phases are functional analogies, with no simulated sleep physiology. Conditional proofs describe persistence, source coverage, partition separation, and matched replay schedules; they do not guarantee semantic truth or learning convergence. A source audit finds that user-role records include machine-generated context, while directory-based project groups can contain related conversations across partitions. We preserve the affected results as historical diagnostics and construct a development comparison with source exclusions, connected-project holdouts, and fixed training budgets. Its main cohorts contain 24 older and 18 newer candidates, with provisional labels and no independent human review. Across three seeds, neither generated condition improves both older and newer classification agreement without greater accepted disagreement against both chronological controls. The learned classifiers remain advisory. A separate sixty-case public retrieval study produced 47 reference-aligned answers, including eleven conditional answers, versus 45 for a direct baseline under unequal call budgets. Neither study establishes improved factual retention or reduced unsupported memories. We specify the additional human judgments, untouched-group evaluation, downstream tests, and resource accounting needed before replay can become a production policy.

1 Introduction

Language agents frequently need information absent from their current context: a user's preference, a correction to an earlier statement, or a detail from a previous task. External memory can retain such information without changing model parameters. Its reliability depends on separate decisions: what to retain, which source supports it, whether a later statement supersedes it, and which evidence to retrieve for a new query.

Summary-based memory introduces an information bottleneck. A fluent summary may omit a detail that later becomes relevant. Improving retrieval over those summaries cannot recover that detail. Conversely, retrieving original conversations exposes the reader to quotations, obsolete statements, and irrelevant requests. A useful memory system needs a compact representation and access to attributed evidence, with explicit lifecycle rules for each.

This work gives a common state model for observations, derived memories, and source excerpts; separates statistical decisions from validated persistence; and implements a private evaluation and training workflow. The empirical study is exploratory. Its purpose is to identify failure mechanisms and establish an auditable baseline. We also separate locally trained classification parameters from external model capabilities, reanalyse class imbalance, and specify the evidence needed for a credible comparison with established memory platforms.

2 Biological motivation and scope

Complementary learning systems theory explains how rapid episodic learning and slower integration can coexist without continually overwriting structured knowledge (McClelland et al., 1995). This motivates separating immutable observations from derived memories. It does not imply that database rows instantiate hippocampal circuits or that summarization is equivalent to synaptic consolidation.

Sleep research supplies more specific hypotheses. Human intracranial recordings during targeted memory reactivation associate successful NREM consolidation with coordinated hippocampal and cortical activity (Duan et al., 2025). In mice, selective disruption of small adult-born neuronal ensembles during REM sleep provides evidence for a causal contribution to associative fear-memory consolidation (Srinivasan et al., 2025). These results concern distinct experimental settings and do not establish a single universal REM consolidation rule.

A computational model by Deperrois et al. (2022) uses complementary waking, perturbed NREM replay, and generative REM phases for visual representation learning. Its dream phase generates novel combinations of prior experience. That study evaluates visual representations; transfer of the mechanism to conversational memory remains an empirical question. Cortex implements attributed capture, explicit offline integration, retrieval, activation decay and optional classifier training. A separate research prototype generates source-grounded and recombined training examples. These phases are software operations with no simulated sleep physiology. The formal results below concern software invariants, not a theorem derived from neuroscience.

Table 1 separates the implemented mechanisms from hypotheses. The word *sleep-inspired* names the architectural motivation; it does not claim biological fidelity. Model inference over a database is not neuronal replay, and relevance decay is not a measurement of synaptic weakening.

Biological motivation	Software interpretation	Status
Rapid episodic capture	Store attributed observations before deriving reusable memories	Implemented
Slower integration	Consolidate observations with explicit source support and contradiction history	Implemented
NREM replay	Revisit source-grounded training examples under a fixed replay budget	Research prototype
REM recombination	Generate recombined training examples, separate from observed history	Research prototype
Selective persistence	Reduce retrieval activation with time and preserve reinforced memories	Engineering heuristic

Table 1: Functional correspondences and their limits. No row establishes a neural mechanism or a simulated sleep stage.

The replay comparison tests whether synthetic variation improves retention of earlier classification behavior and transfer to later conversations. Source attribution and correction tests must accompany those measurements, because a plausible synthetic recollection is not evidence that an event occurred. Generated examples remain separate training artifacts and are never inserted as observed events. Equal update counts and tensor sizes control part of the learning budget; generation and review consume additional resources.

3 Related work

Cognitive Architectures for Language Agents describes modular memory and decision procedures within an agent architecture (Sumers et al., 2024). MemGPT uses hierarchical memory and

control flow to manage a context window (Packer et al., 2023). These works motivate treating memory as more than a collection of vector embeddings. Cortex concentrates on persistence and evidence. Scheduling, user authorization, and the interaction loop belong to its host.

LongMemEval separates indexing, retrieval, and reading, and evaluates information extraction, multi-session reasoning, temporal reasoning, updates, and abstention (Wu et al., 2025). This separation guides our experiments: answer failure alone does not identify the responsible component. Our small public probe is not comparable to a complete benchmark evaluation.

Sentence-BERT supplies a practical representation for semantic similarity (Reimers and Gurevych, 2019). SetFit adapts a sentence encoder through contrastive learning before fitting a classification head (Tunstall et al., 2022). We use these methods as baselines for classifying source spans. Temperature scaling is selected on a development partition to examine probability calibration (Guo et al., 2017).

Memory-R1 learns memory operations using downstream rewards (Yan et al., 2025). MemRL adapts episodic-memory utility without updating the reasoning model's weights (Zhang et al., 2026). Both directions make feedback quality consequential. We keep outcome utility, supervised labels, and model weights distinct. Grading discrepancies in our public probe provide a reason to audit rewards before optimizing a memory policy against them.

Self-Synthesized Rehearsal generates and refines rehearsal examples, then selects a diverse subset for later learning (Huang et al., 2024). Meta-Experience Replay treats transfer and interference as a coupled optimization problem (Riemer et al., 2019). These studies motivate measuring both old-task retention and new-task acquisition. Their results do not establish a benefit for our classifier, source distribution, or memory policy.

4 Problem formulation

An observation is a tuple

$$o = (i, u, x, t_o, \rho),$$

where i is its identifier, u its owner scope, x immutable source text, t_o observation time, and ρ provenance such as speaker and conversation identity. Observation time does not establish the date of an event described in x.

A derived memory is

$$m = (j, u, s, k, c, S, t_\ell, n, z),$$

where s is a summary, k a semantic type, $c \in [0, 1]$ evidence confidence, S a nonempty set of supporting observation identifiers, t_ℓ the last supporting observation time, n a reinforcement count, and z lifecycle metadata. The supported types are fact, behavior, preference, relationship, project, and routine. They describe an annotation policy; they are not mutually exclusive properties of natural language in general.

An atomic candidate is an exact span $x[a : b]$ with half-open Unicode code-point offsets. Atomicity is an annotation goal, not a property guaranteed by span validation. Retention belongs to

$$\mathcal{R} = \{\text{durable}, \text{temporary}, \text{ignore}, \text{uncertain}\}.$$

Retention and type answer different questions. A one-time project request can be temporary. A quotation can resemble a preference without supporting an inference about the user's preference.

For query q and evidence budget B, a conceptual objective is

$$\max_C \mathbb{E}[\text{answer quality} \mid q, C] \quad \text{subject to} \quad \text{tokens}(C) \le B, \quad C \subseteq \mathcal{E}_u,$$

where $\mathcal{E}_u$ is currently eligible evidence for owner u. Rank fusion approximates selection; scope and lifecycle constraints are enforced separately. Quality must distinguish unsupported retention, omitted facts, stale answers, attribution mistakes, and reader errors.

5 Memory operations and retrieval

5.1 Consolidation

Consolidation processes a bounded snapshot of pending observations. A model proposes insertion, reinforcement, or contradiction operations. Validation requires known operation types, bounded summaries, finite confidence values, sources within the current chunk, and update targets in the same owner scope. Contradiction records a successor while preserving the earlier memory's provenance.

Accepted operations and acknowledgement of their exact source snapshot commit in one transaction. Inference occurs outside that transaction. A malformed successful response may receive one format-repair request; the result passes the same validation. This bounds recovery without treating provider calls as exactly-once events.

Inference receives short request-local identifiers for observations and existing memories. An injective mapping π associates each permitted stored identifier with a local alias; the application resolves π^{-1} before validating or committing a proposal. Aliases are chosen outside the set of real identifiers to avoid collisions, and duplicate sources after resolution are rejected. Historical source lists are retained in storage and omitted from the model context, since they cannot support a new operation. These measures reduce identifier-copying burden without expanding the set of permitted sources.

The host decides when consolidation runs and whether observations may be sent to an embedding provider or classifier. Ordinary retrieval does not create a training label.

5.2 Summary and source retrieval

A query can retrieve derived memories or source excerpts. Lexical overlap and cosine similarity produce ranked lists within each representation. Reciprocal rank fusion assigns

$$R(d) = \sum_{c \in \mathcal{C}} \frac{\mathbf{1}\{d \text{ appears in } c\}}{\kappa + \operatorname{rank}_c(d)}, \qquad \kappa = 60,$$

following Cormack et al. (2009). The score orders candidates; it is not a relevance probability.

The default segmentation divides source text into overlapping windows of $W = 512$ code points with overlap $O = 128$. An optional tokenizer-aware segmentation limits both code points and encoder tokens; dense text may require less overlap. Windows retain parent identifiers, offsets, attribution, and observation time. Stored vectors include encoder identity and source hash. Changes to weights or preprocessing require a new model identity.

Code-point limits do not imply token limits. The model card documents a 256-wordpiece truncation default (Sentence Transformers, 2026). Our local adapter uses the embedding tokenizer, including special tokens, to check each window against its 256-token budget. It reduces window length and, when needed, overlap until the next window advances. The embedding endpoint checks the limit independently and rejects oversized inputs. Model and segmentation identities include hashes of weights, tokenizer, and preprocessing policy. A remote tokenizer remains a trusted component: the library validates its declared counts and offsets but cannot independently establish their accuracy. Queries receive the same input-limit check.

An observation supporting any superseded, archived, or expired memory is excluded from source retrieval. This conservative rule can suppress useful information when a source supports

several memories. Unassociated observations can still contain obsolete claims, so eligibility alone does not establish current truth.

5.3 Evidence collection and event dates

A bounded evidence operation combines up to four caller-supplied query variants and deduplicates source windows by parent identifier and offset. Reciprocal rank fusion orders the combined results. The result states whether candidate lists were truncated, whether vector indexing was incomplete, and whether source changes invalidated retrieved material. Its coverage claim is limited to retrieved evidence. Several retrieved sources may describe one event, and a complete ranking list need not contain every relevant event.

Event-date resolution takes an exact source span and an explicit IANA timezone. Supported expressions include a complete calendar date, today, yesterday, a stated number of days ago, and the previous named weekday. Relative expressions use the local civil date of the source timestamp. Dates lacking a year and ambiguous week expressions remain unresolved. The result retains the source hash, original phrase, anchor and resolution rule. This validates date arithmetic, not event occurrence: a quoted or hypothetical event still needs semantic interpretation. The host owns event extraction, identity, correction handling and any final count.

5.4 A source-checked evidence reader

An optional reader uses three bounded model calls: question-based query planning, evidence extraction, and answer composition. One malformed response or inexact quotation may receive a single additional repair call, whose result passes the original validation. The host supplies an explicit question date and timezone and authorizes transfer of selected source excerpts. The reader fuses the existing retrieval channel with a lexical channel that normalizes simple plural endings and month names. Speaker preference and per-source limits reduce crowding by long generic responses. The serialized evidence array is bounded to 32,000 UTF-8 bytes and 80 excerpts. This is a wire-size budget, not a tokenizer guarantee or an exhaustive search claim.

An extracted fact retains an exact source quotation, source identifier, code-point offsets, source hash, observation time and a model-written interpretation. Quotations must occur within the excerpts actually supplied. Proposed date phrases must occur inside those quotations to support arithmetic; an unmatched phrase remains unresolved. Supported missing-year forms use the observation year under an explicitly reported assumption. This optional reader policy is separate from the stricter date-resolution operation, which leaves missing years unresolved.

Civil-day and completed-calendar-month differences are computed deterministically from resolved endpoints. Event counting counts the distinct event identifiers selected by the model, not source rows. The model still decides whether two accounts describe the same event, whether a visit occurred, and whether it falls inside the requested period. Exact quotes establish provenance, not semantic truth. The final answer cites ledger identifiers; its phrasing and interpretation remain statistical outputs. Cited date assumptions and inferred years produce a conditional response status, with assumptions retained separately.

All selected sources are rechecked for owner, content and lifecycle eligibility after composition. If any selected source changed or became unavailable, the operation returns an error without partial facts or an answer. This cannot retract evidence that an authorized provider already received or an answer returned before a later deletion.

5.5 Activation and outcome utility

For an unpinned memory with base half-life $H > 0$, define

$$A(m,t) = 2^{-d/(H(1+\log n'))}, \quad d = \max(0, t - t_\ell), \quad n' = \max(1, n).$$

Time reduces activation while leaving evidence confidence unchanged. Pinning or a disabled half-life sets activation to one. Explicit expiry takes precedence over pinning.

Outcome utility is separate. Starting from $U_0 = 1/2$, an explicitly used memory with a known binary outcome updates as

$$U_{N+1} = \frac{(N+2)U_N + r_{N+1}}{N+3} = \frac{1 + \sum_{i=1}^{N+1} r_i}{N+3}.$$

Duplicate events do not update utility again. The statistic records attributed outcomes; it does not establish that a memory caused success or update model weights.

6 Selective classification

The classifier estimates retention and type distributions for a candidate and its surrounding text. Training and serving use the same input format. A single exact occurrence of the candidate in its context is replaced with a marker to avoid repetition. The complete formatted input is counted with the encoder tokenizer; an oversized input is rejected before inference, without silently removing context. Its training loss is

$$\mathcal{L}(\theta) = -\sum_i \log p_\theta(r_i \mid x_i) - \sum_{i:k_i \text{ annotated}} \log p_\theta(k_i \mid x_i).$$

For the frozen and contrastive baselines, θ denotes linear heads conditional on an encoder; contrastive adaptation precedes head fitting. In the replay experiment, θ includes the encoder and both heads, which are updated jointly with class-weighted cross-entropy. The type loss excludes ignored or unresolved candidates without meaningful type labels. Neither objective learns conflict resolution or generates summaries.

6.1 Trained components and runtime responsibilities

The base representation is the public `all-MiniLM-L6-v2` sentence encoder, which maps text to 384-dimensional vectors (Sentence Transformers, 2026). Cortex does not train a foundation model. The task-specific parameters are two local multinomial logistic-regression heads and, in the adapted variant, a fine-tuned copy of the encoder. For normalized representation $h(x)$, head $j \in \{r, k\}$ returns

$$p_j(y \mid x) = \text{softmax}((W_j h(x) + b_j)/T_j)_y.$$

The historical frozen variant fits only W_j, b_j. Its adapted counterpart first performs eighty SetFit contrastive steps using type annotations, then fits the same heads. Logistic regression uses $C = 10$, seed 17, and at most 1,500 iterations. Each temperature T_j is selected by development loss from $\{0.5, 0.75, 1, 1.5, 2, 3, 5\}$. The replay comparison instead fixes $T_j = 1$ and the final update for every condition. Its joint optimization is described below. Previously examined test partitions are retained as historical diagnostics, not reused as fresh validation.

The retention labels encode persistence policy; the type labels describe content. Neither head extracts candidate boundaries, resolves whether two mentions refer to the same event, or decides whether a new statement contradicts a prior one. Those capabilities must be evaluated separately. The external language model still proposes consolidation operations; local classifiers have no authority over those writes.

Component	Function	Learned in this study
External language model	Propose annotations, consolidation operations, and answers	No weight updates
MiniLM encoder	Represent candidate text and context	Frozen or locally adapted
Retention head	Durable, temporary, ignore, or uncertain	Locally fitted parameters
Memory-type head	Fact, behavior, preference, relationship, project, or routine	Locally fitted parameters
Persistence validator	Check source identity, scope, and allowed transitions	Deterministic code

Table 2: Training and runtime roles. The pretrained encoder and external language model are third-party components. Task-specific heads and adapted weights are experimental local artifacts, not released model checkpoints.

The decision rule abstains if maximum retention probability is below 0.8, the chosen retention is uncertain, or a retained candidate's maximum type probability is below 0.8. These are experimental operating points. High probability on synthetic text does not establish calibration on real conversations.

Records retain source spans, grouping identifiers, label provenance, and annotation-policy version. Human annotations, designed synthetic labels, and model-generated labels remain distinguishable. The loader rejects exact normalized candidate text across partitions and refuses provisional teacher labels unless explicitly enabled. This validation cannot establish that a user-role record was written by a human or that two directory-derived groups are independent. The source and grouping audits below address these separate assumptions.

Temperature is chosen on development negative log likelihood. Test labels do not participate in fitting or temperature selection. Metrics include macro-averaged F1, multiclass Brier score, calibration error, coverage, and accepted-prediction error. Exact deduplication does not eliminate semantic leakage among templated examples.

The operational boundary remains observational: predictions are returned to the caller and never modify memory state. A host can inspect disagreement before a learned decision suppresses useful information or retains an unsupported assertion.

7 Source-grounded and recombined replay

The research cycle separates experience capture, training-example generation, model updates, and evaluation. Captured observations retain source identity and attribution. A candidate extracted from an observation remains an exact span. A generated paraphrase or recombination belongs to a synthetic training namespace, with explicit contributing source identifiers; it is never imported as a new event in the memory store. External generation and review models receive authorized training sources only. The production classification interface remains advisory.

7.1 Source and partition audits

The initial inventory treated selected user-role records as human input. Inspection of source metadata and known message envelopes found automatic goal reminders, compaction summaries, and wrappers containing both an assistant's question and a human answer. The audit excludes these records as a whole. Recovering an answer from a mixed wrapper would require attributed extraction and new labels. Pattern-based exclusions do not certify every remaining record as human-authored.

Directory names also failed as independent project identifiers. A session can move among sev-

eral related repositories. We therefore construct a graph with vertices for source sessions, original group identifiers, and known repositories in working-directory metadata. Edges connect a session to its groups and repositories. Connected components define a conservative related-project partition. Every generated derivative inherits all contributing source components. Metadata alone cannot identify every semantic relation, but it exposes overlaps hidden by the original grouping.

Before fitting the corrected comparison, we reserve the largest formerly train-only component containing both older and newer candidates, using counts and a fixed hash tie-break. Its complete source lineage is excluded from training. Original development components absent from the training set join the corresponding evaluation cohorts. Examples from components shared with training are evaluated separately as related-project diagnostics. The resulting independent development cohorts contain 24 older candidates in three components and 18 newer candidates in four components, with one component shared across time. Their labels were examined in prior work, so they are not fresh confirmatory data.

7.2 Paired replay construction

Let $e_i = (x_i, r_i, k_i)$ denote an earlier training candidate with its provisional retention and optional type labels. For each primary e_i, generation proposes a meaning-preserving paraphrase p_i and a recombination g_i with a second eligible source. A paraphrase must preserve entities, attribution, negation, intent, temporal scope, and certainty. A recombination may change the activity or setting while preserving the primary decision structure and using a meaningful concept from the second source. An incompatible pair may produce no recombination. The contributing sources can belong to the same connected project component.

Two models label each generated utterance without the intended labels or generation condition. Generation uses Luna 6; annotation uses Luna 6 and Sol 6.1; a separate Sol 6.1 request checks faithfulness, coherent decision scope, and actual source contribution. Medium effort is requested through the local API, with returned model identifiers and usage retained. Acceptance requires both annotations to match the primary retention/type labels, atomicity, and a successful semantic check. The semantic reviewer shares a model with one annotator, and the generator also supplies an annotation. These correlated model judgments neither replace human review nor establish that a generated statement occurred.

Local validation checks source membership, recorded ancestry, text duplication, and the complete 256-token encoder budget. Both conditions must pass for a primary to enter the paired library. One source appears once as a primary. The minimum of 64 unique primaries from four independent source components was fixed before fitting. An initial library fell below that requirement after auditing. A declared extension attempted every remaining eligible primary once, with a deterministic secondary preferring compatible provisional labels. The full batch was completed without stopping when the minimum was reached. Rejected outputs and failed calls remain part of resource accounting.

7.3 Joint adaptation and controls

The encoder and two linear heads are trained jointly. For batch B, retention labels r_i, type labels k_i, and typed subset $B_k = \{i \in B : k_i \text{ is annotated}\}$, the loss is

$$\mathcal{L}_B = \frac{\sum_{i \in B} w_{r_i}[-\log p_r(r_i \mid x_i)]}{\sum_{i \in B} w_{r_i}} + \frac{\sum_{i \in B_k} v_{k_i}[-\log p_k(k_i \mid x_i)]}{\sum_{i \in B_k} v_{k_i}}.$$

The second term is omitted when its denominator is zero. Weights are inverse square roots of class counts in the initial training pool, normalized to mean one within each head and fixed

thereafter. Normalized embeddings are scaled by 20 before the linear heads. The exported head coefficients absorb that scale.

Each seed begins with 64 updates containing 16 earlier natural candidates and 16 original synthetic candidates. The second stage has 64 updates of 32 examples. All conditions share the same 16 newer candidates at each update. The other 16 are sampled from the complete earlier pool for ordinary mixed training, or from a common paired-source schedule for repeated examples, paraphrases, and recombinations. The paired arms therefore share source identity and the exact label sequence. Ordinary mixed training retains its broader source distribution.

The temporal cutoff is 12 October 2026 at 02:14:29.962 UTC. It was set from the original natural training inventory before the source audit and retained during regrouping. Source timestamps determine temporal membership; they do not establish when a described event occurred.

AdamW uses encoder learning rate $2 \cdot 10^{-5}$, head learning rate 10^{-3}, weight decay 0.01, and global gradient clipping at one. Every batch is padded to 256 tokens without truncation. Seeds are 17, 41, and 73. Conditions share the seed-specific initial checkpoint and begin their second stage with a fresh optimizer. The final checkpoint is fixed in advance; temperature is one and the existing joint acceptance threshold is 0.8. There is no checkpoint selection or temperature fitting within this comparison.

A pooled ordinary reference uses the exact multiset of 4,096 presentations from the corresponding initial and ordinary mixed stages, shuffled once with a fixed seed offset. It starts from the same base initialization and resets the optimizer halfway through its 128 updates. It has access to later examples from the first update, so it is an offline reference with no valid pre/post forgetting interpretation.

Example presentations, update counts, capacity, and training tensor dimensions are controlled. Actual non-padding tokens, generation and review inference, and wall-clock contention differ. Equal tensor dimensions are not a measurement of equal hardware work or equal end-to-end cost.

7.4 Outcomes and interpretation

The primary outcomes are retention/type agreement on the component-disjoint older and newer cohorts, change from the initial checkpoint, and joint acceptance coverage and disagreement. Reports retain all class supports, seed results, and separately measured synthetic and related-project cohorts. Retention macro-F1 averages all four declared classes, assigning zero to unsupported classes; supported type macro-F1 averages only types present in the cohort. Class support and constant-label references are therefore necessary to interpret the absolute scores. With three seeds and few independent components, paired seed intervals are descriptive and cannot certify population-level transfer.

Here, retention of performance means preserving classification decisions on older examples. Factual knowledge retention, source-correct downstream answers, and unsupported-memory creation require separate tests. A classifier can improve its macro-F1 while causing a complete memory system to omit a needed fact. A replay mechanism becomes a production candidate only if it improves older-knowledge retention and unseen-conversation performance against ordinary and repeated-example controls without increasing unsupported memories.

Proposition 1 (Recorded-lineage separation)**.** *Let $c(o)$ map each source record to its connected component. Suppose training and evaluation use disjoint component sets C_T, C_E. For any admitted training example z, let $A(z)$ be its complete recorded source ancestry and require $\{c(o) : o \in A(z)\} \subseteq C_T$. Then no source in an evaluation component is a recorded ancestor of an admitted training example.*

Proof. If an evaluation source o were an ancestor of admitted z, admission would imply $c(o) \in$

C_T, while evaluation membership would imply $c(o) \in C_E$. This contradicts $C_T \cap C_E = \varnothing$. For multiple generation stages, the same argument applies to transitive ancestry, provided every stage preserves the complete ancestor set. □

The property is conditional on correct ancestry and grouping. It does not rule out unrecorded semantic overlap or foundation-model pretraining exposure.

Proposition 2 (Matched label schedules)**.** *Suppose repeated, paraphrased, and recombined examples at each scheduled position share the same retention label, optional type label, and source index. If all arms receive the same newer-example positions, then their per-update retention counts, type counts, and type-label masks are identical. If all batches also use shape* 32×256*, their padded-token counts are identical.*

Proof. Equality of labels and masks holds position by position for the paired examples by assumption and for the newer examples by the common schedule. Summing their indicator functions over each batch gives equal class counts. Each update contains $32 \cdot 256$ padded positions, so summing over equal update counts gives equal padded-token totals. □

The proposition makes no claim that gradients, real token counts, generation costs, or generalization are equal. It does not establish that replay improves learning.

8 Conditional properties

The following results concern an abstract state machine. They assume atomic transactions, immutable observations, compliant writers, and correct owner authorization by the host. They are consequences of those assumptions, not a machine-checked proof that every implementation path satisfies them.

Proposition 3 (Scope preservation)**.** *If initial memories reference only observations in their own owner scope, and all source additions and update targets are checked against that scope, every reachable memory preserves the property.*

Proof. Induct on committed transitions. Insertion establishes the property through source validation. Contradiction creates a checked successor and a same-owner predecessor link. Reinforcement unions the existing source set with checked sources. Other transitions remove records or change metadata without adding sources. Rejected transactions leave the preceding state unchanged. □

Proposition 4 (Atomic acknowledgement)**.** *If every consolidation writer acquires the same per-owner lock before selecting its pending snapshot, retains that lock through commit, and commits operations and acknowledgements atomically, no observation is acknowledged by two successful chunks.*

Proof. The lock orders successful consolidations for an owner. At the first committed acknowledgement, the observation ceases to be pending and later snapshots exclude it. A failed transaction rolls back both operations and acknowledgements. No second successful chunk can therefore select that observation as pending. □

This result permits several facts to cite one observation and permits repeated inference after failure.

Proposition 5 (Exact-replay exclusion)**.** *Suppose imports and deletion share an owner-scoped barrier. Deletion atomically writes a content tombstone and removes the source, and subsequent imports check tombstones. Once deletion commits, importing identical source text for that owner cannot restore it.*

Proof. The barrier orders an import before or after deletion. An import ordered before deletion is part of the source state examined and removed. An import ordered after deletion sees the committed tombstone and skips insertion. Identical source text cannot survive either ordering. □

Fingerprints are assumed collision resistant. The property excludes paraphrases, copies already returned, provider retention, and separately trained weights. In-flight reads are governed by their final eligibility snapshot; deletion cannot retract an output already authorized by an earlier snapshot.

Proposition 6 (Activation monotonicity)**.** *For fixed $n' \geq 1$ and $H > 0$, activation is nonincreasing in elapsed time. For fixed positive elapsed time it is nondecreasing in n'.*

Proof. For $d > 0$,

$$\frac{\partial A}{\partial d} = -\frac{\log 2}{H(1+\log n')} A < 0, \qquad \frac{\partial A}{\partial n'} = \frac{d \log 2}{H n'(1+\log n')^2} A > 0.$$

For $d \geq 0$, the exponential formula gives $0 < A \leq 1$ and continuity at zero. Together with the derivatives, this establishes the stated weak inequalities. □

An older assertion is not necessarily less likely to be true.

Theorem 7 (Window coverage)**.** *Let source length $L > 0$, width W, and overlap $0 \leq O < W$ be integers. Generate windows starting at $j(W - O)$, stopping at the first whose end reaches L. They cover the source. Every contiguous span of at most $O + 1$ code points lies wholly in at least one window.*

Proof. Consecutive starts differ by $S = W - O \leq W$, so windows have no gap. The first starts at zero and the last reaches L. For a span starting at a, choose the latest generated start $jS \leq a$. If it is the last start, its window contains the span because it reaches L. Otherwise $a < (j+1)S$, hence $a \leq jS + S - 1$. For length $\ell \leq O + 1$,

$$a + \ell \leq jS + S - 1 + O + 1 = jS + W.$$

Thus the whole span fits. □

Coverage does not guarantee that embedding preserves meaning or retrieval selects the window.

Proposition 8 (Validated token-window coverage)**.** *Let accepted windows be intervals $[a_i, b_i)$ over a source of length $L > 0$, with $a_1 = 0$, $b_n = L$, $a_{i+1} \leq b_i$, and strictly increasing starts and ends. Their union is $[0, L)$. If a correct tokenizer reports at most T tokens for each window, and the encoder uses that tokenizer and special-token policy, no accepted window exceeds T encoder tokens.*

Proof. The first interval covers the source prefix ending at b_1. Inductively, the next interval begins at or before the covered prefix ends and extends it to b_{i+1}. The final prefix therefore ends at L. The token statement follows from equality between the reported count and the encoder's count for each interval. □

This result guarantees complete source coverage, not containment of every multi-character fact in a single window. Variable overlap does not inherit the fixed-overlap theorem's $O+1$ span guarantee.

Proposition 9 (Civil-date subtraction)**.** *Represent each valid Gregorian calendar date as midnight UTC. Dividing the difference in their Unix-second representations by* 86,400 *gives their signed difference in civil days, independent of daylight-saving transitions in the timezone used to resolve the dates.*

Proof. UTC midnights in this representation are separated by exactly 86,400 Unix seconds for consecutive civil dates. Summing these equal differences between the two dates gives the claimed integer. The source timezone affects which civil dates are chosen, but not subtraction of those dates. □

The representation does not model leap seconds or elapsed local hours. It is used for civil-day questions only.

Theorem 10 (Selective-risk bound)**.** *Fix a classifier and acceptance rule before testing. Assume independent test draws from the deployment distribution with correct labels. Conditional on $m > 0$ accepted examples, let $\widehat{R}$ be their mean zero-one error and R the population error conditional on acceptance. For $\delta \in (0, 1)$, conditional on that accepted count, with probability at least $1 - \delta$,*

$$R \leq \min\left\{1, \widehat{R} + \sqrt{\frac{\log(1/\delta)}{2m}}\right\}.$$

Proof. For a fixed rule, accepted examples are independent draws from the distribution conditional on acceptance. Their error indicators lie in $[0, 1]$ with mean R. Hoeffding's inequality gives $\mathbb{P}(R - \widehat{R} \geq \epsilon \mid m) \leq e^{-2m\epsilon^2}$ (Hoeffding, 1963). Substitute $\epsilon = \sqrt{\log(1/\delta)/(2m)}$ and cap at one. □

Selecting the rule using these test labels invalidates this experiment. The bound cannot certify deployment risk from templated synthetic data or unreviewed teacher labels.

Proposition 11 (Source-bound answer publication)**.** *Assume source hashes are collision-free on the compared records. Suppose every published fact is checked against a supplied source excerpt, selected sources retain their content hashes, and all selected sources pass an owner and eligibility check immediately before publication. Then every quoted fact in a successful answer is a substring of a source that was eligible at that final check.*

Proof. Quote validation establishes membership in a supplied excerpt and records its parent source hash and offsets. The final check establishes that the same source content remains eligible in the requested scope. Substring membership is preserved by content equality. If either check fails, no answer is published, so a successful output has the stated property. □

The result is snapshot-relative. It neither verifies the model-written interpretation nor eliminates the interval between the final database read and delivery to a caller.

9 Experimental design

The studies use authorized local conversation records and a sample from LongMemEval. Historical local partitions were assigned by directory-derived groups before fitting. Subsequent audits found machine-authored records and related projects across those partitions. We retain

their measured values with that qualification. The replay comparison uses corrected source exclusions and connected-project separation; it remains developmental because these labels have been examined before.

For public evaluation, twelve previously examined cases form the development set. Sixty additional cases are selected by a fixed hash ordering, with ten from each question type and no overlap with the nineteen cases used in earlier work. Each case receives a separate owner scope and its full supplied conversation history. Questions and reference answers never enter the memory store. Both readers use the same evidence-size ceiling, but their model-call budgets differ. The requested model is Luna 6 at medium effort through the local API. Date interpretation uses an explicit UTC reference. Scoring checks consistency with the supplied history and reference; it does not independently verify the truth of historical assistant recommendations. Model judgments are followed by Codex inspection, with no independent human adjudicator.

Classification uses the pinned MiniLM sentence encoder and the same candidate-context format for training and serving. Frozen and contrastive baselines and their original temperature selection are reported historically. The chronological comparison trains the encoder and heads jointly from fresh checkpoints, with fixed temperature, update count, and acceptance rule. Synthetic examples and natural candidates are reported separately. All natural annotations remain provisional model labels.

The local execution environment uses PostgreSQL 17 in Docker, Go 1.26.10 and a tokenizer-checked ONNX embedding adapter. A private four-slot serving variant accelerates the larger evaluation; its vectors and segmentation were checked against the serial adapter, including concurrent repeated requests. Shared machine and provider load prevents interpreting timings as an isolated throughput benchmark. Source versions, budgets and aggregate measurements are recorded in the accompanying manifest.

10 Results

10.1 Source and related-project audit

The source audit excluded 168 messages containing known machine or mixed-authorship material. In the 3,585-example dual-model dataset, 116 accepted natural candidates were affected: 35 training, 76 development, and five previously tested candidates. The 1,592 original synthetic examples are separate. The remaining 1,877 natural candidates retain provisional labels.

Forty of 123 inspected sessions named more than one known repository in their working-directory metadata. Conservative grouping found 16 connected components. One component contained 1,700 natural candidates across all three original partitions. The original directory-based partitions therefore cannot substantiate independent-project transfer. An additional inventory of the same file-based stores contained 79 new message identifiers but no new independent connected component. These additional messages were reserved outside the present fit. This inventory did not exhaust every local application store; subsequent collection from additional history databases and desktop session stores is separate from this frozen experiment.

The audit also traced the earlier 3,032-candidate corpus used in the historical tables below. It found 176 affected authorship candidates, including nine in that corpus's test partition, and 2,760 candidates in a connected component spanning all original partitions. These are different corpora and denominators. Historical metrics are preserved; removing contaminated evaluation rows alone would not repair affected training.

10.2 Corrected replay comparison

The final library contains 75 matched primary sources from four independent source components. Its retention distribution is 57 temporary, six durable, eleven ignored, and one uncertain candidate in each paired arm. The typed primaries include 53 projects, five facts, three preferences, one routine, and one behavior; no relationship example survived. These imbalances limit any class-specific conclusion.

The fitting pools contain 317 earlier and 168 newer natural candidates, plus 1,063 original synthetic training examples. The main development evaluation has 24 older and 18 newer candidates, with 22 and 18 typed candidates respectively. Related-project diagnostics contain another 269 older and 851 newer candidates; they are never counted as independent transfer. The synthetic development cohort contains 262 examples.

The older cohort contains 19 durable, three temporary, and two ignored candidates. Its types are 18 routines, two facts, and two projects. The newer cohort contains 14 durable and four temporary candidates, with nine routines, six preferences, and three projects. No uncertain label appears in either cohort. These small, uneven supports leave several classes unevaluated and limit the interpretation of macro-F1. Full aggregate confusion matrices, calibration measures, and related-project diagnostics accompany the manuscript.

The two contributing generation batches attempted 625 source pairs and produced 1,119 structurally valid utterances. Thirty-two pairs failed generation, 67 returned no recombination, and 105 invalid generation responses were recorded across attempts. One annotation batch failed, leaving sixteen examples without complete review. Model agreement and semantic checks retained 123 paired sources before local source, duplicate, and connected-partition checks. Recorded generation and review responses returned 1,399,449 tokens, including rejected and invalid outputs. Usage for attempts without a returned usage record is unknown. A verified monetary charge was not returned.

Condition	Old F1	Change	New F1	Type F1	Coverage	Error
Ordinary mixed	0.411	0.078	0.202	0.786	0.481	0.331
Repeated examples	0.346	0.014	0.271	0.500	0.463	0.333
Paraphrase replay	0.223	-0.109	0.114	0.500	0.481	0.562
Recombined replay	0.338	0.006	0.197	0.510	0.500	0.667
Pooled ordinary	0.674	–	0.369	0.545	0.463	0.151

Table 3: Means across three seeds on component-disjoint development cohorts. Old and new retention F1 average all four declared classes; type F1 averages supported types in the newer cohort. Change is relative to the seed-specific initial checkpoint. Coverage and error concern the joint acceptance rule. Pooled ordinary has no chronological change score. These are agreements with provisional labels, not factual-retention or unsupported-memory measurements.

Neither generated condition simultaneously improves mean older and newer retention agreement without greater mean accepted disagreement against both chronological controls. This comparison does not support promoting a generated replay policy. The audit verified identical newer-example schedules, shared initial checkpoints, paired source and label schedules, and 524,288 padded positions per adaptation stage. Each pooled reference matched the corresponding chronological ordinary arm's complete presentation multiset and 1,048,576 padded positions. Real token counts and provider inference remain separate resources. The training-majority retention label is temporary; its older and newer macro-F1 scores are 0.056 and 0.091. The training-majority type is project, with supported macro-F1 0.095 on newer examples. These uncalibrated constant predictions are agreement references, not acceptance policies.

Classification agreement does not measure unsupported autobiographical memories. No human-reviewed labels have been imported, and no classifier is promoted to a memory-writing

Condition	Seed	Old retention F1	New retention F1
Ordinary mixed	17	0.317	0.071
Repeated examples	17	0.410	0.321
Paraphrase replay	17	0.056	0.071
Recombined replay	17	0.410	0.321
Ordinary mixed	41	0.338	0.071
Repeated examples	41	0.191	0.172
Paraphrase replay	41	0.338	0.071
Recombined replay	41	0.522	0.199
Ordinary mixed	73	0.577	0.464
Repeated examples	73	0.438	0.321
Paraphrase replay	73	0.277	0.199
Recombined replay	73	0.083	0.071

Table 4: All chronological run outcomes. Seeds share evaluation examples and are not independent new conversation samples.

policy. The earlier variable-padding experiment is preserved as a qualified diagnostic: recombination improved mean older-cohort macro-F1 from 0.532 to 0.547 relative to ordinary mixed training, while newer-cohort macro-F1 fell from 0.481 to 0.408 and accepted disagreement rose from 17.5% to 25.8%. That experiment used affected source records and overlapping related projects; it cannot establish a replay advantage.

10.3 Serving and persistence checks

A predetermined seed-17 recombination artifact was exported solely for interface verification, without selecting the highest-scoring run. On 32 candidate fixtures, the Python service and current Go API reproduced saved probabilities with a maximum absolute difference of $6.78 \cdot 10^{-7}$. Source hashes and the abstention rule matched. Twelve invalid-request or source-lifecycle checks returned rejection, covering owner mismatch, deleted and archived evidence, invalid spans, incompatible model or policy identifiers, empty requests, duplicate identifiers, and input limits. The isolated PostgreSQL store contained zero memories and zero training events before and after classification. These checks verify the advisory contract and do not measure semantic memory quality.

The current software commit also passed 86 Go tests with the race detector and PostgreSQL 17, with no skipped tests, plus static analysis and builds. Both database checks used disposable local containers. No private observation or generated training example was written to a production store.

10.4 Historical source and reader diagnostics

The following measurements predate the authorship and related-project audits. Their numeric values are retained; the natural classifier partition does not establish independent-project transfer. The separate public question-answer study is not affected by the local directory grouping error.

10.5 Historical natural-conversation study

A broader inventory inspected 640 local session files. After removing machine context, generated subagent messages, detected secret patterns and exact normalized duplicates, the original filter retained 2,814 user-role records from 119 sessions and 35 directory-derived groups: 1,336 records

from Codex and 1,478 from Claude. The audit in Section 10.1 subsequently found machine-authored records and related projects across partitions. These messages contained 2,111,697 characters. All eligible messages were submitted for provisional annotation; long messages were split explicitly rather than truncated. Annotation failures were retried in smaller batches. One retry batch still failed and supplied no accepted labels.

The teacher produced 3,235 candidate spans. Validation excluded 192 invalid annotations, ten exact candidate duplicates across partitions, and one duplicate identifier. The remaining 3,032 candidates all fit the complete encoder input budget. They contained 1,763 temporary, 535 durable, 443 ignored and 291 uncertain labels. These are policy annotations by a model, with no human-reviewed labels.

A fixed hash ordering assigned 21 directory-derived groups to training, seven to development and seven to testing, before classifier fitting. This produced 1,629, 1,040 and 363 natural candidates respectively. Synthetic training and development partitions were added as in the earlier study. The following table uses only the natural test partition; 310 candidates have a type label.

Training	Retention	Type	Accepted	Errors
Synthetic only, frozen	73/363	186/310	108/363	78/108
Mixed, frozen	215/363	279/310	61/363	18/61
Mixed, adapted	233/363	285/310	45/363	13/45

Table 5: Historical agreement with provisional labels on seven directory-derived groups. The later audit found related-project overlap and affected source authorship. The joint decision rule uses the fixed 0.8 thresholds. An accepted error is an incorrect retention label or an incorrect type for a retained candidate.

Natural examples improved agreement relative to synthetic-only training, but this comparison does not establish useful performance against simple constant-label policies. The adapted model reached 64.2% retention agreement and 91.9% type agreement. It accepted only 12.4% of candidates, with 28.9% disagreement among accepted predictions. This is insufficient evidence for autonomous retention decisions. The apparent zero-error result among seven accepted predictions in the smaller study did not persist at this scale. A live Go API check reproduced offline predictions for 361 candidates with matching context, with a maximum probability difference below 10^{-6} and no memory writes. Two candidates at an explicitly split source boundary had different complete context and were excluded from this numerical-equivalence check.

10.6 Class imbalance and calibration

Table 6 reports the natural test support and the adapted model's correct predictions. The dataset reflects one owner's work conversations: 253 of 363 retention labels are temporary, and 286 of 310 typed candidates are projects. Always predicting temporary would agree on 69.7% of retention examples; always predicting project would agree on 92.3% of typed examples. These are retrospective constant-label accuracy checks, not trained models or independently validated policies. Both exceed the adapted model's corresponding accuracy. The adapted type head predicts project for 309 of 310 inputs and preference for one; all correct type predictions are projects.

The type macro-F1 of .160 and zero recall on every supported non-project type qualify the 91.9% headline accuracy. No six-type generalization claim follows from this test. The retention head also recovers only one of 37 durable examples. Although development calibration reduces aggregate probability error relative to synthetic-only training, it does not make high-confidence retention reliable. More provisional examples alone do not remedy an ambiguous annotation policy or missing class coverage. Human review, class-specific errors, and representative held-out

Head	Provisional label	Test support	Correct	Recall (%)
Retention	Durable	37	1	2.7
	Temporary	253	213	84.2
	Ignore	32	16	50.0
	Uncertain	41	3	7.3
Type	Fact	9	0	0.0
	Behavior	0	0	—
	Preference	14	0	0.0
	Relationship	0	0	—
	Project	286	285	99.7
	Routine	1	0	0.0

Table 6: Class support and adapted-model recall against provisional teacher labels. Missing test classes are unevaluated. Counts are calculated from the frozen confusion matrices, not a new inference run.

	Macro-F1		Brier		ECE	
Model	Ret.	Type	Ret.	Type	Ret.	Type
Synthetic, frozen	.133	.153	1.336	.576	.648	.178
Mixed, frozen	.339	.158	.547	.169	.072	.052
Mixed, adapted	.341	.160	.510	.147	.074	.031

Table 7: Additional metrics from the same frozen predictions. Macro-F1 includes all four retention and all six declared type labels, assigning zero to undefined class scores; two type classes have no test support. Brier is multiclass squared probability error. ECE uses ten equal-width confidence bins. Lower Brier and ECE are better, but imbalance can make either look favorable while minority classes fail.

groups are required before assigning either head operational authority.

10.7 Operational and answer evaluation

The full natural corpus indexed into 7,184 source windows. An observed whitespace-only window exposed a client-side rejection that had interrupted indexing; accepting nonempty whitespace preserved complete coverage. The final local indexing run completed without silently truncating an input.

Large-context consolidation also exposed invalid model-generated source references. Validation rejected those proposals and retained completed batches. The final context uses request-local aliases and omits historical source lists. Aliases resolve to exact IDs before the existing transaction checks. The operational results below distinguish successful acknowledgement from semantic correctness of the retained memories.

The recovered full-corpus run acknowledged all 2,814 observations exactly once, with no missing or foreign source references. A fresh run using local aliases processed 2,600 observations before a summary-length rejection; a retry processed the remaining 214. Across the fresh run and retry, fifteen successful chunks acknowledged all observations, and one rejected chunk acknowledged none. The original recovery run contained five rejected chunks. These tests establish observed persistence behavior, not the factual accuracy of every retained memory.

The twelve-case development set was evaluated twice with the frozen reader. Source review found 11 reference-aligned answers in the first run and 12 in the second, including respectively three and four explicitly conditional answers. The first run abstained on an ambiguous museum-visit date. In the second, it provided the reference value under a stated interpretation of that

date. Thus 12/12 was attained in one run; the combined result was 23/24, not evidence of perfect repeatability.

10.8 Held-out public evaluation

Sixty additional LongMemEval cases were selected before implementation changes: ten per question type, ordered by a fixed SHA256 rule, excluding all nineteen cases used in earlier development. Full supplied histories were ingested under separate owner scopes. Questions and reference answers were excluded from memory ingestion. The reader implementation was frozen before the final measurement. Earlier partial executions were retained privately and were not used to select prompts or tune against reference answers.

Both conditions received at most eighty excerpts and 32,000 serialized UTF-8 evidence bytes. The baseline used single-query retrieval and one direct reading call. The source-checked reader used query planning, extraction and composition, with at most one additional format repair. This is an equal evidence-size comparison, not an equal-call-budget comparison. Scoring combined model grading with Codex review of references and source evidence. No independent human judge participated. Table 8 reports qualified answers separately.

Reader	Complete	Conditional	Partial	Incorrect	Error
Direct baseline	44	1	4	11	0
Source-checked	36	11	4	6	3

Table 8: Sixty held-out cases per condition. A conditional answer states an assumption and gives the matching reference value. Such an answer does not establish that the assumption is true. Reference alignment combines the first two columns; it is distinct from unqualified correctness.

Question type	Direct baseline	Source-checked
Knowledge updates	8/10	9/10
Multi-session reasoning	6/10	7/10
Assistant recall	10/10	8/10
Personalized responses	3/10	6/10
User facts	10/10	10/10
Temporal reasoning	8/10	7/10

Table 9: Reference-aligned answers, including conditional answers, out of ten cases per type. Partial answers and errors are excluded.

The source-checked reader returned 47/60 reference-aligned answers (78.3%), compared with 45/60 (75.0%) for the baseline. Complete, unconditional answers accounted for 36/60 (60.0%) and 44/60 (73.3%), respectively. The alignment measure includes the eleven and one conditional answers shown separately in Table 8. The two-case difference is too small to establish a reliable advantage. Personalization improved, while several assistant-recall and temporal cases deteriorated. The reader produced more conditional answers and three validation errors. Two errors rejected inexact quotations; one rejected an unresolved arithmetic endpoint. Counts of distinct events and retrieval of every required event remain weaknesses.

The final source review also found a timezone presentation inconsistency: date arithmetic used the configured timezone, while a displayed source timestamp could use the database driver's timezone. The release normalizes the model's anchors to the requested timezone. It also permits one inexact quotation to use the existing repair allowance while preserving exact-match validation. A single literal today or yesterday phrase can supply a conditional date candidate when the ex-

tractor omitted an anchor; this does not establish the event date. These post-measurement fixes are covered by regression checks and targeted replay; they are not credited with changing the frozen sixty-case score. A further release-model pass matched eleven of the twelve development references, missing a location detail. The earlier 12/12 result was therefore not stable across repeated model calls.

The larger natural-label study and the public study address different tasks. Improved agreement with provisional retention labels does not establish improved answer quality, and reference agreement does not certify truth outside the supplied conversation. The learned classifiers remain advisory.

These conditions measure retrieval and reading with access to original source messages. They do not isolate the fidelity of consolidated memories or show that compression preserves every fact needed for future questions. Equal evidence bytes also do not imply equal model tokens, latency, or inference cost. A consolidated-memory-only condition and matched reader budgets are needed to attribute any gain to the memory representation itself.

11 Discussion

The diagnostics distinguish separate constraints. Source access can recover information omitted by summaries, but a narrow budget can still exclude a required event. Short excerpts may improve budget utilization while separating related context across windows. Classification narrows a model's decision task, but performance on designed examples does not establish transfer to natural conversations.

Observation time, event time, and validity time have different meanings. Explicit date resolution addresses arithmetic and preserves the source anchor. It does not extract every event or identify repeated descriptions of one event. Multi-session counting still requires semantic event identity and correction handling. The present classifier performs none of those operations.

Deletion exposes a boundary between external memory and learned parameters. Persistence can remove source records and derived memories. Once private text contributes to weight updates, source deletion does not itself remove that information from weights. Training artifacts remain private, and weight-training experiments use fabricated examples and, separately, owner-authorized provisional annotations. Production training needs separate consent, minimization, model-retention, and retraining policies.

Exact vector scans and explicit capacity bounds simplify small-scale operation. They do not establish suitability for large corpora. Authentication, retention policy, scheduling, and provider access remain host responsibilities. Learned mutation policies would have to preserve these boundaries.

12 Platform context and limits of comparison

Published platform results provide reference points for the next evaluation, not a ranking against this study. Table 10 records vendor-reported LongMemEval headline scores inspected on 22 October 2026 (UTC). We did not rerun these systems. Their source scope, readers, inference budgets, and grading differ from the sixty-case Cortex experiment.

Mem0 describes additive temporal metadata and time-sensitive retrieval ranking (Mem0, 2026b). Zep's Graphiti represents changing entity relationships in a temporal knowledge graph (Zep, 2026a). Hindsight describes semantic, keyword, graph, and temporal retrieval, followed by reranking, as well as a separate reflection operation (Vectorize, 2026). These are concrete design comparisons. Their benefit for a particular workload still requires an experiment.

Platform	Reported (%)	Reported scope and protocol qualification
Mem0	94.4	Managed platform, top-200 retrieval; mean 6,787 context tokens. Proprietary optimizations are excluded from the open-source SDK (Mem0, 2026a).
Zep	90.2	451/500 questions; multiple retrieval scopes and a cross-encoder reranker. Reader and judge are GPT-5.4; reader uses medium reasoning (Zep, 2026b).
Hindsight	94.6	LongMemEval-S headline on the project overview. Its recall and agentic reflection paths have different call budgets; the headline alone does not identify a matched configuration (Vectorize, 2026).

Table 10: External, vendor-reported results, unverified by this study. Cortex's 78.3% reference alignment includes conditional answers on a different sixty-case sample and is deliberately excluded from this table. No percentage-point gap or superiority claim can be inferred.

Cortex emphasizes a small Go/PostgreSQL package with explicit source support and state transitions. It has lexical and semantic retrieval, contradiction history, activation decay, and source-linked date arithmetic. It does not yet have a validated learned retention policy, a general event-identity model, or an evaluated graph-based retrieval stage. Its engineering checks establish observed behavior under tests; the current evidence does not establish competitive answer quality, lower cost, or lower latency than the platforms above.

13 Limitations

The proofs apply to the stated abstract machine. Regression tests give evidence of implementation conformance in selected cases, not exhaustive verification. No theorem establishes that an extracted statement is true, a summary complete, or a probability calibrated.

The private corpus belongs to one owner and cannot be publicly reproduced. Pattern checks can accept incorrect qualifications. User-role metadata did not establish human authorship, and changing working directories obscured related projects. The corrected development cohorts are small, and source exclusions recognize known templates rather than certifying every remaining author. The original seven-case and twelve-case public probes influenced development. The sixty-case public measurement is historical; it is no longer untouched validation for subsequent changes. Synthetic wrapper splits share topics and vocabulary. All provisional labels and semantic judgments await independent human review. These constraints require exploratory reporting and continued advisory operation.

14 Protocol for comparative evaluation

This section specifies the remaining confirmatory work. The developmental replay comparison above does not complete this protocol or a platform benchmark. Benchmark readiness means that the runs are controlled and reproducible; competitive capability requires favorable results from those runs.

14.1 Freeze the test contract and separate prior exposure

The first public suite should be the full 500-question LongMemEval-S release, with dataset checksum, case identifiers, source-role policy, and question timestamps fixed before running (Wu et al., 2025; LongMemEval authors, 2025). The twelve-case development set stays a regression suite. The nineteen earlier cases and sixty-case study have already influenced development or

review; their union of 79 cases must be flagged. Report full-suite results for comparability and the remaining 421 cases separately as the previously unexamined subset. This separation does not establish that a foundation model never saw the public benchmark in pretraining.

Pin each implementation commit, package version, model identifier, embedding model, prompt, tokenizer, inference setting, and evaluator configuration. Archive a public protocol before inspecting new answers. Any change made after a failure creates a new experimental condition. Confirm that source ingestion excludes benchmark questions, answers, and evaluator-only evidence flags. A subsequent LoCoMo evaluation should use its official data and category definitions as a distinct transfer test (LoCoMo authors, 2024); do not merge its score with LongMemEval.

14.2 Compare memory representations and complete products

Run two tracks. In the controlled-retrieval track, each system supplies evidence to the same frozen reader. In the complete-product track, each platform uses its documented answering path, with every call and cost recorded. If a managed service cannot expose or fix an internal model, report that limitation and keep it out of claims about equal-model performance. A self-hosted Graphiti run is not a reproduction of the managed Zep headline; the same distinction applies to Mem0's open-source SDK and managed platform.

Condition	Question it isolates
Raw-history retrieval	What does a fixed lexical/dense baseline recover without consolidation?
Cortex memories only	What survives the consolidation bottleneck?
Cortex source windows only	What does attributable raw evidence recover?
Cortex combined retrieval	Does combining summaries and sources improve over either alone?
Classifier off versus on	Does a learned retention decision improve utility or discard required evidence?
Pinned competitor versions	How does the same reader perform on each retriever's evidence?
Oracle evidence sessions	How much error remains when the reader receives the benchmark's relevant sessions?

Table 11: Proposed controlled conditions. Use identical questions, source access, and reader settings where the condition permits them. Oracle evidence is a labeled diagnostic and is never available to ordinary ingestion or retrieval.

Use a common tokenizer and an 8,000-token evidence ceiling for the first controlled comparison. State truncation and packing rules in advance. Compare one-reader-call conditions separately from a bounded, four-call condition; do not grant extra planning calls to only one system. Run three answer repetitions per question at fixed settings, recording seeds where supported. Repeated answers measure inference variability and are not independent new questions. Charge ingestion, consolidation, embedding, reranking, and answer calls separately; report cold and warm p50/p95 latency, failures, and cost amortized over an explicit number of queries. Report bytes and tokens separately.

14.3 Measure correctness, evidence, and operational behavior

Preserve the official benchmark score using a pinned evaluator, and additionally report complete, conditional, partial, unsupported, erroneous, and correctly abstained answers under a documented human-review rubric. Do not silently count a conditional answer as established fact. Two reviewers should independently assess a predeclared stratified sample and all disputed cases, with disagreement and adjudication recorded. Reviewers should be blinded to the system identity.

Measure evidence recall against available source annotations, citation precision, answer correctness by question type, and consolidated-memory compression at a fixed downstream accuracy. Attribute failures to ingestion, consolidation, retrieval, source interpretation, or answer generation when the evidence permits. Include corrected facts, repeated event mentions, event counting, changing preferences, missing dates, and timezones. Use paired confidence intervals for answer differences; resample at the shared history or source-group level where questions are correlated. Fix the primary comparison and any multiple-comparison correction before scoring. Report inconclusive differences as inconclusive.

Keep lifecycle tests separate from semantic accuracy: retry and crash recovery, cross-owner isolation, exact-source deletion and reimport, source invalidation during an answer, context overflow, and incomplete indexing. A model-quality gain cannot compensate for losing source ownership or acknowledging an uncommitted write. Publish protocol, versions, aggregate metrics, and permitted public outputs. Private transcripts, labels, and trained weights remain outside the public repository.

14.4 Establish labels before promoting a classifier

Write an annotation guide that distinguishes persistence policy from content type, direct user claims from quotations or requests, and observation time from event time. Collect independently human-reviewed examples across consenting owners and source groups, with all four retention labels and all six types represented. Keep a naturally distributed test set and a separate balanced diagnostic set; report both and mark unsupported classes unevaluated. Include constant-label and frozen-encoder baselines. Test groups and owners must be excluded from fitting, replay generation, threshold selection, and calibration.

Choose confidence thresholds on development data. Before giving a classifier write authority, require a predeclared upper bound on accepted-prediction error, useful coverage, and acceptable class-specific false retention and false deletion. For illustration, a 5% error ceiling at 95% confidence under the stated Hoeffding bound requires at least 600 error-free accepted independent test examples. Correlated examples from one conversation do not satisfy that independence assumption. This is a proposed risk target, not a certification of the current classifier. Cases outside the validated policy remain advisory or return to human review.

14.5 Test the sleep analogy as an ablation

Confirm the developmental comparison of direct training, source-grounded replay, matched duplicate-example training, and recombined synthetic replay on untouched groups. Hold presentations, update steps, model capacity, and selection budget fixed, and match paired label schedules. Report real and padded tokens and additional generation resources. Audit authorship and connected-project ancestry before partitioning. Generate every replay example exclusively from training components. Preserve links to contributing sources and identify generated examples as synthetic; never insert a recombination as an observed user event.

Measure retention and type macro-F1, selective risk and coverage, performance on old and newly introduced domains, and downstream answer quality on untouched groups. A replay gain must exceed the matched-compute control and survive correction and source-attribution tests. This experiment can support or reject the computational analogy; it cannot establish that the implementation reproduces NREM or REM physiology.

14.6 Criteria for a comparative claim

A reproducible report must identify every tested condition, retain failure denominators, and publish uncertainty alongside accuracy, cost, and latency. A claim of superiority needs a predeclared

paired comparison whose interval supports improvement without violating lifecycle constraints. A claim of comparable quality needs a non-inferiority margin chosen before the run and a useful cost or operational benefit. Neither criterion is met by matching a vendor's headline percentage on a different dataset slice. Until these conditions are satisfied, Cortex should be described as an experimental memory engine with tested persistence behavior and an explicit route to comparative evaluation.

15 Conclusion

Cortex applies a sleep-inspired separation between rapid source capture and slower integration to an inspectable memory engine. Its implemented contribution is source-preserving storage, bounded retrieval, and validated transitions. The formal results are conditional software properties; they establish neither neural fidelity nor semantic truth.

The replay prototype makes source-grounded rehearsal and controlled recombination executable hypotheses. The source audit also shows why additional training runs alone are insufficient: authorship, related-project grouping, and label quality determine what a measured gain means. Source-assisted answering has not shown a reliable advantage over its baseline. Production adoption requires simultaneous improvements in older-knowledge retention and unseen-conversation performance, with no increase in unsupported memories, against ordinary and repeated-example controls. Independent human judgments, fresh confirmation, and downstream tests remain necessary. No proof in this paper guarantees that the learning process will converge to that outcome.

A Implementation and reproducibility

The accompanying manifest records implementation identity, environment, encoder revision, and artifact hashes. These identify an experiment; they do not define the scientific claims. Public source includes classification policy, training and labeling tools, and regression tests. Private conversations, per-example predictions, provider responses, database exports, and trained artifacts are excluded.

A complete observation's source windows publish in one transaction after source eligibility is checked. Retrieval checks eligibility again after provider work. The service assumes a trusted host and binds to localhost by default. Exposed deployments must supply authentication and owner authorization.

Two offsetting grading discrepancies left an earlier public total unchanged: an unsupported abstention passed for a preference question, while a response containing the correct degree subject failed because it qualified whether graduation was established. The case-level correction shows why aggregate agreement alone is inadequate.

B Earlier diagnostic studies

The following pilot studies establish the earlier failure analysis and use smaller, separately described samples. Their totals are not pooled with the expanded study.

B.1 Synthetic classification

Encoder	Retention F1	Type F1	Coverage	Error
Frozen	0.731	1.000	0.926	0.050
Contrastively adapted	0.943	1.000	0.944	0.000

Table 12: Synthetic test results. Retention uses 108 examples; type uses the 84 with type annotations. Coverage and error concern the retention head at a probability threshold of 0.8, not the joint API abstention rule.

The frozen model misclassified all 12 ignored examples in the held-out wrapper family as durable. Contrastive adaptation reduced this count to four. The adapted model abstained on those errors at the reported retention threshold. Zero observed accepted errors in this synthetic sample is not a deployment guarantee: the sampling assumptions of the selective-risk theorem are not established.

Retention Brier score was 0.148 for the frozen model and 0.032 for the adapted model. Type F1 was 1.000 for both models, but the vocabulary and facts were shared across wrapper families. These measurements demonstrate successful fitting of a designed task, not reliable classification of arbitrary conversations.

B.2 Transfer to provisional session annotations

The private review queue contains 84 teacher-proposed spans from the 51 authorized messages. No label in this queue has been independently reviewed by a human. The teacher labeled 66 spans temporary, nine uncertain, five ignored, and four durable. Thus the provisional distribution differs sharply from the synthetic test partition, which contains 72 durable and 12 examples of each other retention label.

Encoder	Retention agreement	Type agreement	Joint abstentions
Frozen	9/84	47/74	51/84
Contrastively adapted	10/84	39/74	61/84

Table 13: Agreement with provisional teacher annotations, including predictions on abstained candidates. Type agreement uses the 74 annotated types. These are agreement measurements, not human-validated accuracy.

Both classifiers ran through the Go shadow API. The improvement on synthetic retention did not transfer to this diagnostic. The low agreement, distribution shift, and uncertain reference labels provide no basis for promoting either classifier to memory control. A reviewed natural-conversation corpus is required before training or evaluating an operational replacement. One of the 84 formatted classification inputs also exceeded the encoder context limit. This finding motivated exact budget validation in the follow-up study.

B.3 Retrieval under a fixed evidence budget

Evidence	Local checks	Public checks	Public complete
Whole observations	19/19	5/7	4/7
Source windows	19/19	5/7	4/7
Memories and windows	19/19	5/7	5/7

Table 14: One reader run per condition. Public checks are automatic; complete answers are judged by a subsequent Codex case inspection, not independent human review.

All conditions passed the 19 local expected-answer checks. Each passed five of seven automatic public checks and failed multi-session counting and elapsed-time reasoning. Inspection marked two passing recommendation answers as partial: they named the preferred editor but did not explicitly address the requested advanced-learning focus. The combined condition did address that focus. This post hoc completeness check illustrates why a keyword pass is insufficient evidence of full answer quality.

The window representation preserved the observed local check count under the fixed budget. These measurements do not establish a general improvement over whole-source retrieval. The temporal responses abstained rather than supplying the earlier unsupported zero-day interval, but they still did not answer the question. Neither smaller excerpts nor the revised reader instruction solved the missing-evidence problem.

The frozen-memory diagnostic preceding this comparison passed 14 of the same 19 local checks using summaries alone. That result motivated source retrieval; it does not isolate a causal effect of the present window design. Replaying all 26 window queries against the final implementation produced identical retrieval outputs.

A tokenizer audit of the 3,563 retained observations found 1,165 full observations longer than the encoder's 256-wordpiece limit. The window representation contained 10,145 excerpts; 20 still exceeded that limit, with a maximum of 427 wordpieces. Thus character windows substantially reduced input-length violations in this sample but did not eliminate truncation. Token-aware segmentation remains necessary for a strict encoder-context guarantee.

B.4 Grading audit and implementation checks

Case inspection changed two of 21 prior public judge decisions. A memory-only answer containing the correct degree subject was accepted, while an unsupported abstention on a personalized recommendation was rejected. These changes left the aggregate memory score at 3/7 and the source and combined scores at 5/7. The audit was performed by Codex against stored questions, reference answers, and reader outputs; it is not independent human adjudication.

Regression checks cover source scope, invalid distributions, exact offsets, deletion during inference, all-or-nothing window publication, and explicit indexing. The database race suite, static checks, build, and source vulnerability scan passed. These checks support specific implementation contracts; they do not measure semantic completeness or general answer quality.

B.5 Historical directory-group follow-up

The follow-up sampled 63 additional explicit user-role messages from ten local project groups (25 messages from Codex and 38 from Claude). The teacher proposed 90 spans. Three retained candidates lacked a type and failed the annotation contract; they were excluded without inventing a label. The remaining 87 spans contained 67 temporary, eight ignored, eight uncertain and four durable labels. All remained provisional, with no human-reviewed examples.

A fixed hash order assigned six directory-derived groups to training, two to development and two to testing before model fitting. This produced 43, 28 and 16 natural candidates respectively. The training and development sets combined these candidates with the corresponding synthetic partitions. The test below contains only the 16 natural candidates from the original test groups. Directory separation alone does not establish independent projects, as the later source audit demonstrates. A synthetic-only frozen encoder supplies a matched baseline using the same input format. The adapted comparison uses 80 contrastive steps. The development partition selects temperature, while the joint acceptance rule remains fixed at 0.8.

Training	Retention	Type	Accepted	Errors
Synthetic only, frozen	2/16	7/14	3/16	2/3
Mixed, frozen	11/16	13/14	6/16	1/6
Mixed, adapted	10/16	13/14	7/16	0/7

Table 15: Historical agreement with provisional annotations on two directory-derived test groups. The first two measures include abstained predictions. Errors among accepted predictions count a wrong retention label or, for retained candidates, a wrong type. These labels are not human-validated truth.

Natural examples improved agreement in this small comparison, while the adapted model accepted fewer than half the test candidates. Zero disagreements among seven accepted predictions cannot certify operational risk. The corpus is dominated by project requests and does not establish performance on other conversational distributions. All models remain advisory. A live service check classified the 16 candidates through the Go API and created no memories. A separate oversized-input check was rejected before inference.

B.6 Frozen retrieval follow-up

The later public probe selects two cases per question type using a fixed SHA256 ordering, excluding the original seven development cases. The twelve supplied histories contain 5,832 turns, of which 5,820 have nonempty text and are indexed. Questions and reference answers were excluded from ingestion; the reader receives only the question and retrieved evidence. Unlike

the initial diagnostic, this experiment indexes source evidence directly without a consolidation pass.

Both retrieval conditions use the exact-tokenizer adapter and a 4,000-token serialized-evidence budget measured with `cl100k_base`. The single-query condition retrieves up to 40 windows. The multi-query condition merges up to four query variants, generated from the question alone by Luna 6 at medium effort, into the same 40-window limit. The reader, also Luna 6 at medium effort, is asked to preserve attribution, apply explicit corrections and enumerate distinct events before counting. These design changes and the larger budget prevent a direct score comparison with the earlier 1,200-token experiment.

Retrieval	Complete	Partial	Incorrect
Single query	9	0	3
Multiple queries	8	1	3

Table 16: Frozen public follow-up, twelve cases per condition. Codex inspected answers against the reference answers; no independent human adjudication was performed. Partial answers are not included in the complete total.

The single-query condition produced 9 complete answers out of 12. Multiple queries produced 8 complete answers, one inconsistent partial answer and three incorrect answers. This small probe provides no evidence of an aggregate improvement from query expansion. One expanded retrieval answered a studio-location question missed by single-query retrieval; another answer incorrectly withheld confirmation of a location despite retrieving its explicit source. A temporal answer stated both four and five months and was not counted as complete. Both conditions abstained on an elapsed-duration question even though dated departure and return evidence was present. These cases distinguish evidence availability from successful use by the reader.

Advice questions were assessed for meaningful personalization to prior context; illustrative suggestions in their reference answers were not treated as mandatory wording. The assessment remains subjective and was performed by Codex, not by an independent human. The small sample, stochastic query generation and reader responses do not establish statistical superiority of either condition.

B.7 Input and implementation checks

The final input audit covered 11,400 stored observations and 32,260 tokenizer-aware windows. Every source was completely covered; no accepted window exceeded the 256-token limit. The maximum observed count was 254. The same sources produced 21 over-budget windows under the fixed 512-code-point scheme. This checks preservation and input limits, not semantic retrieval quality.

All 19 prior private recall pattern checks passed with the strict adapter. These are repeated development diagnostics, not new independent questions. The final source binary reproduced all 24 public-condition evidence contexts exactly, separating documentation changes from the measured retrieval implementation. PostgreSQL race tests, Python contract tests, static checks, builds, vulnerability checks, and full-history secret scanning passed. Live checks confirmed strict rejection of an oversized classifier input, source-bound timezone resolution, and no memory writes from classification.

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